CSE221: Algorithms

Chap	Course Contents	Chapter Notes	Resources
0	Introduction to Algorithms	CSE221 Chap 0	
1	Asymptotic Analysis - Big Oh, O - Big Omega, Ω - Big Theta, θ	CSE221 Chap 1	
2	Time Complexity - Simplified Rules - Series Rules - Iterative Time Complexities (dependent, independent, nested loops) - Recursive Time Complexities (Substitution, Master theorem, Recurrence Tree)	CSE221 Chap 2	[Note] Advanced Master Theorem [BuX] Time Complexity Part 1 [BuX] Time Complexity Part 2 [BuX] Time Complexity Part 3 [BuX] Time Complexity Part 4
3	Searching Algorithms - Linear Search - Binary Search - Ternary Search	CSE221 Chap 3	[Recording] Ternary Search
4	Sorting Algorithms - Merge Sort - Quick Sort - Count Sort	CSE221 Chap 4	[Recording] Merge and Quick Sort
5	Divide-and-Conquer - Maximum Subarray Problem - Karatsuba's Multiplication Algorithm	CSE221 Chap 5	[YouTube] Maximum Subarray (Watch till 6:11) [YouTube] Karatsuba's Multiplication [Recording] Karatsuba's Multiplication
	MID EXAMINATION		
7	Graph Traversal 1 - BFS - BFS Applications (Shortest path & Bipartite)	CSE221 Chap 7	[Recording] BFS and BFS Applications [Recording] DFS and DFS Applications
7	Graph Traversal 2 - DFS - DFS Applications (Edge Classfication & Cycle Detection)	CSE221 Chap 7	[Recording] BFS and BFS Applications [Recording] DFS and DFS Applications
8	Graph Algorithms - Topological Sort - Strongly Connected Components - Shortest Path (Dijkstra)	CSE221 Chap 8	[YouTube] Dijkstra's Algorithm [YouTube] Bellman-Ford Algorithm
9	Greedy Algorithms - Spanning & Minimum Spanning Tree - Kruskal's Algorithm - Huffman Coding	CSE221 Chap 9	[BuX] Huffman Coding [Recording] Kruskal's Algorithm
10	Dynamic Programming - DP Basics - Knapsack Problem (Fractional & 0/1) - Longest Common Subsequence	CSE221 Chap 10	[YouTube] Intro to DP [YouTube] 0/1 Knapsack using DP [YouTube] LCS
11	Backtracking Algorithms - 8 Queen Problem - 15 Puzzle Problem	CSE221 Chap 11	